United States Geological Survey

Reference Material Information Sheet

BCR-2 (Columbia River Basalt)

Introduction

Rock used in the preparation of BCR-2 was collected in 1996 from the Bridal Veil Flow Quarry under the direction of Stephen A. Wilson, U.S. Geological Survey. The quarry is located approximately 26 miles east of Portland, Oregon and is the same collection site used to provide material for BCR-1 (Flanagan, 1967, 1969).

The USGS has not published a metrologically traceable characterization of the elemental content of this material. The values presented in Table 1 and Table 2 are from a published study that summarizes multi-laboratory analyses of mass fraction values for elements in the material. It is recommended that the user of this material consult the references provided to determine the estimated value(s) that are the best fit for their purpose.

Expiry

The material is considered geochemically stable with proper storage

Contact

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Table 1

Estimates of major oxide concentrations in BCR-2 from published literature

Oxide	Mass Fraction (g/100 g)	Uncertainty (g/100 g)	n[1]	Notes
Al_2O_3	13.48	0.12	*20*	*2,4*
CaO	7.114	0.075	*22*	*2,4*
Fe_2O_3T	13.77	0.19	*20*	*2,4,5*
K_2O	1.774	0.019	*20*	*2,4*
MgO	3.599	0.044	*20*	*2,4*
MnO	0.1966	0.0030	*21*	*2,4*
Na_2O	3.120	0.042	*20*	*2,4*
P_2O_5	0.3593	0.0095	*17*	*2,4*
SiO_2	54.00	0.20	*19*	*2,4*
TiO_2	2.265	0.024	*30*	*2,4*

Table 2

Estimates of trace element concentrations in BCR-2 from published literature

Element	Mass Fraction (mg/kg)	Uncertainty (mg/kg)	n[1]	Notes
Ba	683.9	4.7	*79*	*3,4*
Be	2.17	0.10	*25*	*3,4,6*
Bi	0.050	0.015	*8*	*3,4,6*
Cd	0.69	0.29	*9*	*3,4,6*
Ce	53.12	0.33	*89*	*3,4*
Co	37.33	0.37	*60*	*3,4*
Cr	15.85	0.38	*58*	*3,4*
Cs	1.160	0.023	*62*	*3,4*
Cu	19.66	0.72	*58*	*3,4*
Dy	6.424	0.055	*78*	*3,4*
Er	3.670	0.038	*78*	*3,4*
Eu	1.989	0.024	*79*	*3,4*
Ga	22.07	0.19	*47*	*3,4*
Gd	6.811	0.078	*78*	*3,4*
Hf	4.972	0.034	*7*	*3,4,7*
Ho	1.313	0.011	*78*	*3,4*
La	25.08	0.16	*79*	*3,4*
Li	9.13	0.22	*36*	*3,4*
Lu	0.5049	0.0078	*7*	*3,4,7*
Mo	250.6	6.7	*21*	*3,4*
Nb	12.44	0.20	*74*	*3,4*
Nd	28.26	0.37	*23*	*3,4,7*
Ni	12.57	0.30	*59*	*3,4*
Pb	10.59	0.17	*74*	*3,4*
Pr	6.827	0.044	*79*	*3,4*
Rb	46.02	0.56	*11*	*3,4,7*
Sb	0.302	0.029	*14*	*3,4,6*
Sc	33.53	0.40	*68*	*3,4*
Sm	6.547	0.047	*20*	*3,4,7*
Sn	2.28	0.13	*22*	*3,4*
Sr	337.4	6.7	*13*	*3,4,7*
Ta	0.785	0.018	*72*	*3,4*
Tb	1.077	0.026	*77*	*3,4*
Th	5.828	0.050	*15*	*3,4,7*
Tl	0.267	0.018	*19*	*3,4,6*
Tm	0.5341	0.0060	*64*	*3,4*
U	1.683	0.017	*16*	*3,4,7*
V	417.6	4.5	*60*	*3,4*
W	0.465	0.050	*13*	*3,4*
Y	36.07	0.37	*73*	*3,4*
Yb	3.392	0.036	*78*	*3,4*
Zn	129.5	1.8	*60*	*3,4*
Zr	186.5	1.5	*75*	*3,4*

Notes for Table 1 and Table 2

The number of decimal places in mass fraction and uncertainty values are as given in the citation.

(1) n is the number of laboratory mean estimates that were used to calculate the overall mean and uncertainty of the estimate reported in the cited publication. See the cited publication for details.
(2) Estimate from Table 3 of Jochum et al., 2016.
(3) Estimate from Table 4 of Jochum et al., 2016.
(4) Uncertainty is expressed as 95% confidence level.
(5) Fe_2O_3T is total iron expressed as Fe_2O_3.
(6) As reported by Jochum et al. (2016), measurements were all made by a single technique (inductively coupled plasma-mass spectrometry).
(7) As reported by Jochum et al. (2016), measurements were made by isotope dilution-mass spectrometry.

References

References provided in addition to those cited may also be of interest to users of this reference material.

Flanagan, F.J., 1967, U.S. Geological Survey silicate rock standards: Geochimica et Cosmochimica Acta, v. 31, no. 3, p. 289-308. [Also available at doi:10.1016/0016-7037(67)90043-9.]

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Govindaraju, K., 1994, 1994 compilation of working values and sample description for 383 geostandards: Geostandards Newsletter, v. 118, p. 1-158. [Also available at https://doi.org/10.1046/j.1365-2494.1998.53202081.x-i1.]

Jochum, K.P., Weis, U., Schwager, B., Stoll, B., Wilson, S.A., Haug, G.H., Meinrat, A.O., and Enzweiler, J., 2016, Reference values following ISO guidelines for frequently requested rock reference materials: Geostandards and Geoanalytical Research, v. 40, p. 333-350. [Also available at https://doi.org/10.1111/j.1751-908X.2015.00392.x.]